

Zubairu Gamidov  
22301571

I affirm that I have not given or received any unauthorized help on this report and that this work is my own

## Low Cost DC-DC Converter

### Hardware Implementation

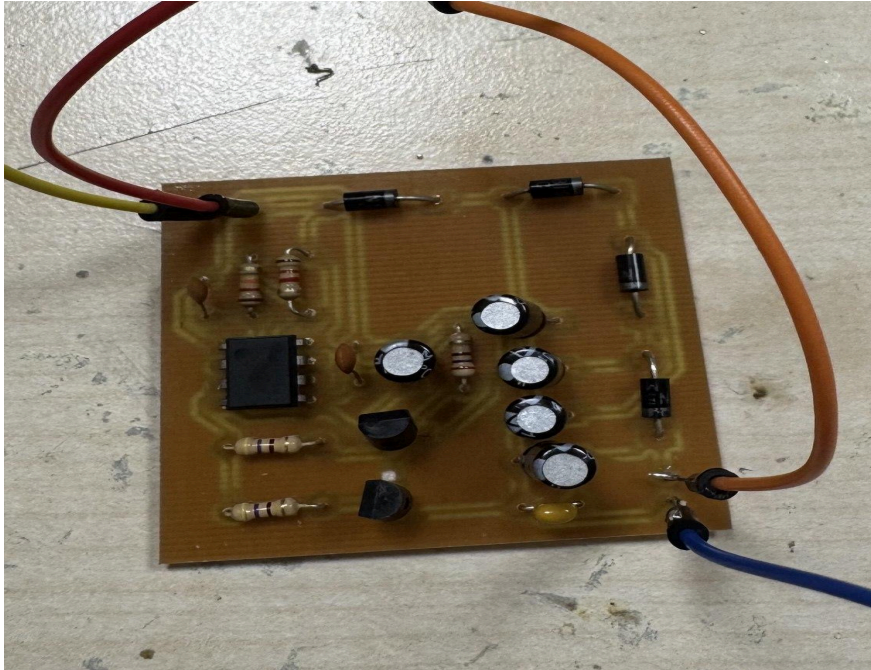


Figure 6. Top Side of a Soldered PCB

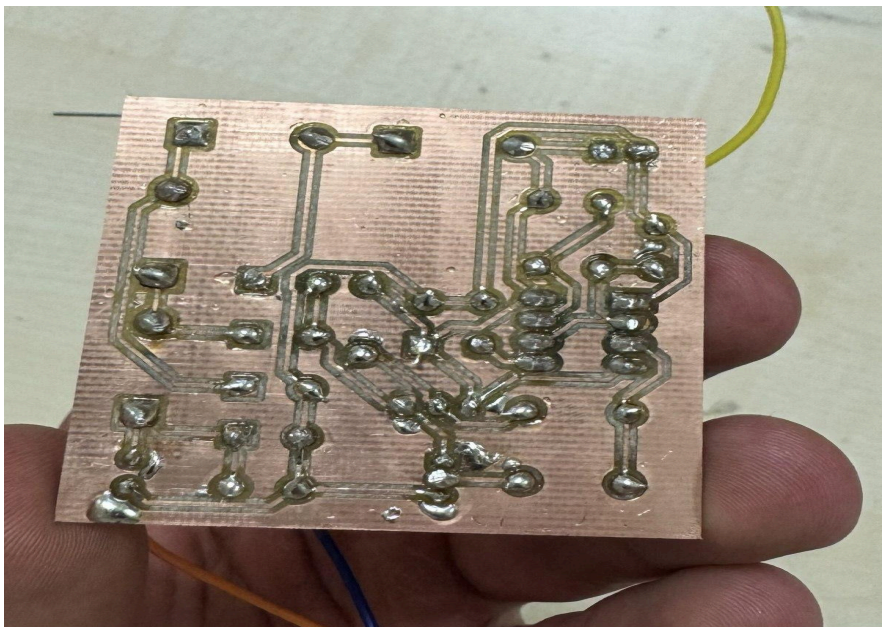


Figure 7. Bottom Side of a Soldered PCB



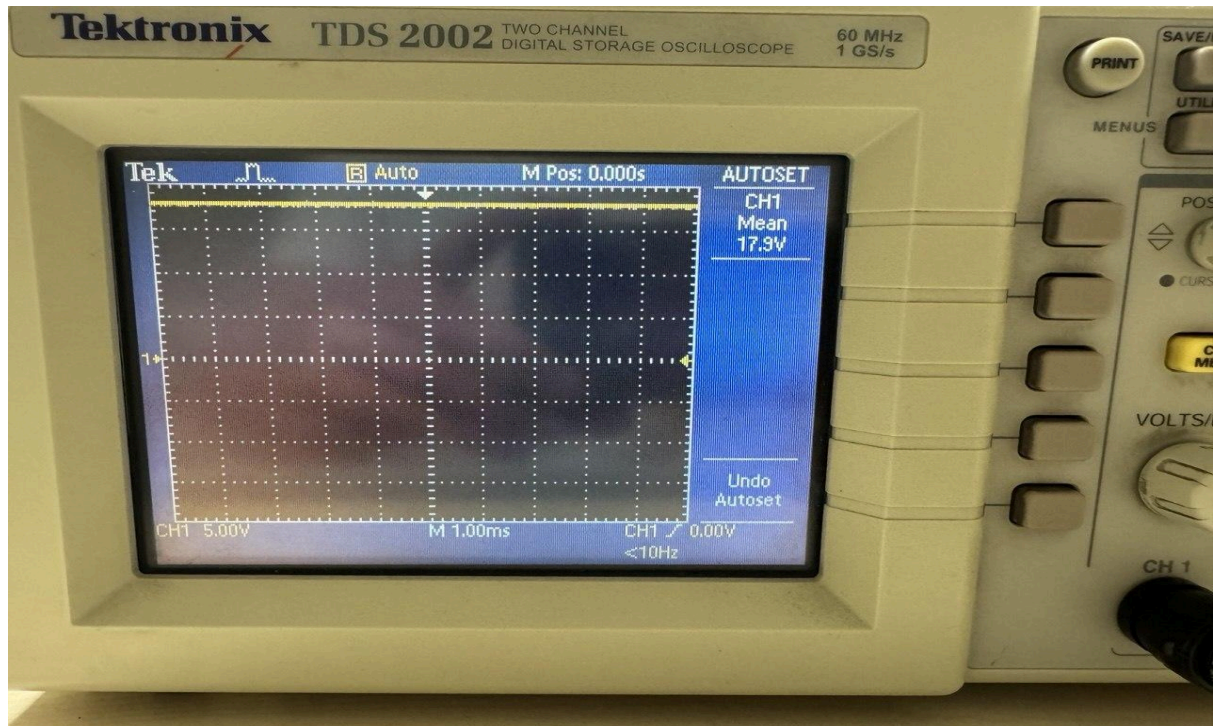


Figure 8.  $V_{out}$  without the load

As we predicted in our design phase on LTspice, we should get 18V or a little less in our output without the load. **17.9V** with no load achieved.

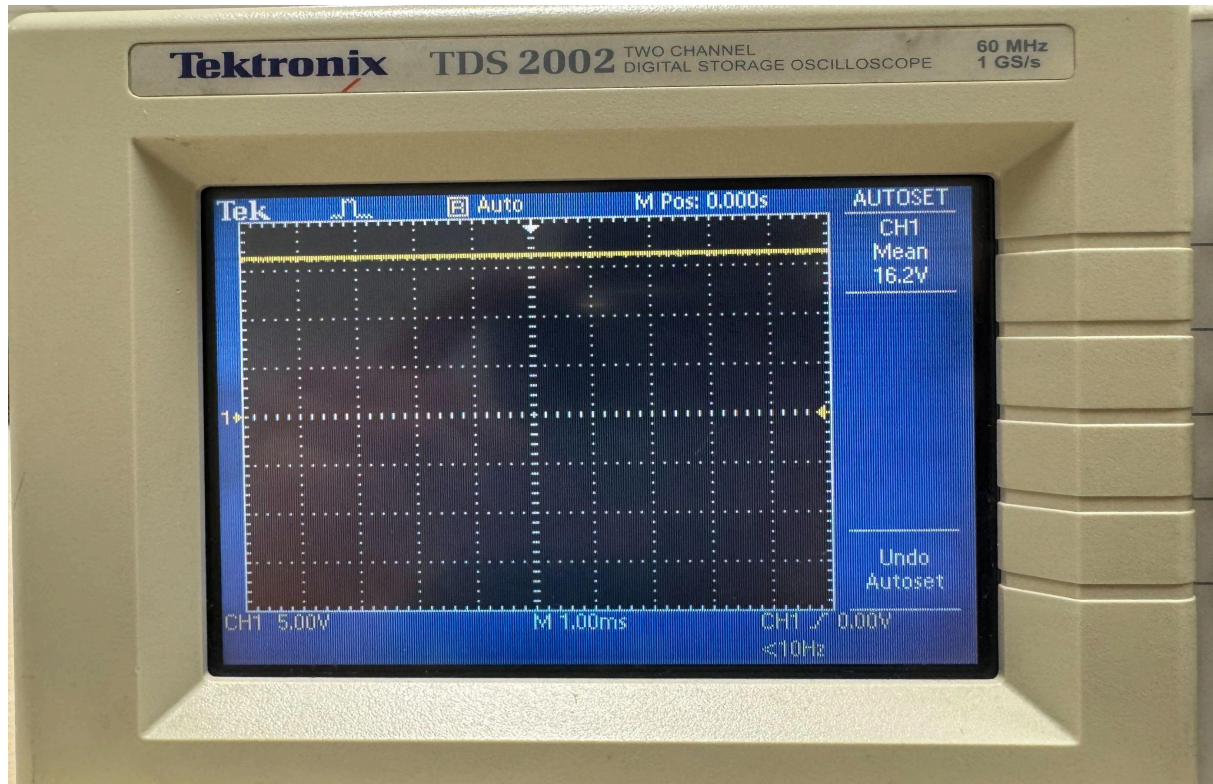


Figure 9.  $V_{out}$  with the load resistor

Stable **16.2V** in the output node, which is enough greater than required 15.0V



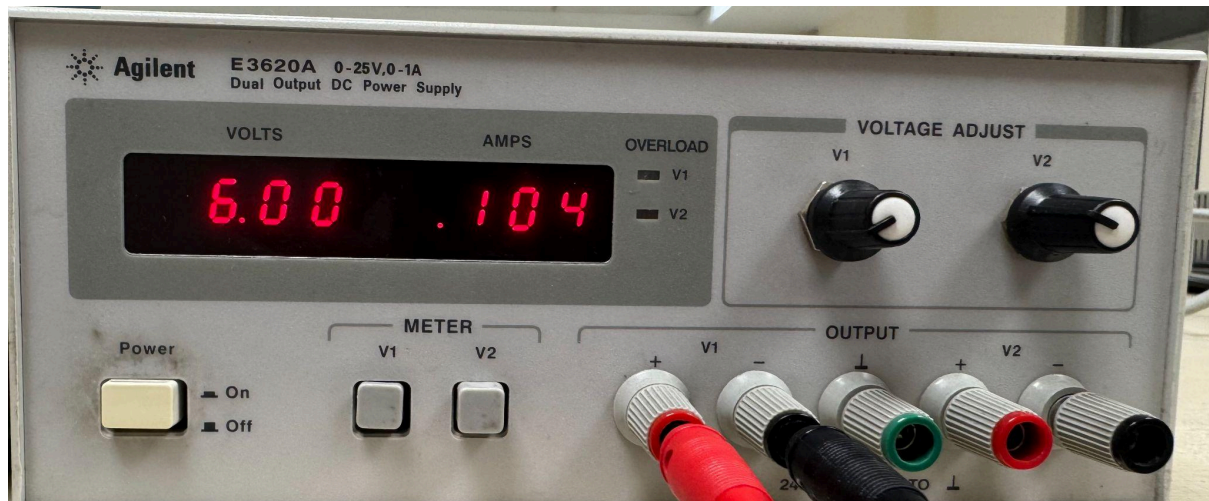


Figure 10. Power supply current with load connected

With 16.0V output, our circuit consumes **104mA** current with 6V input.

Following calculations gives us an efficiency:

$$\eta = \frac{V_{out}^2}{I_s * R_L * V_{in}} = 75.1\%.$$

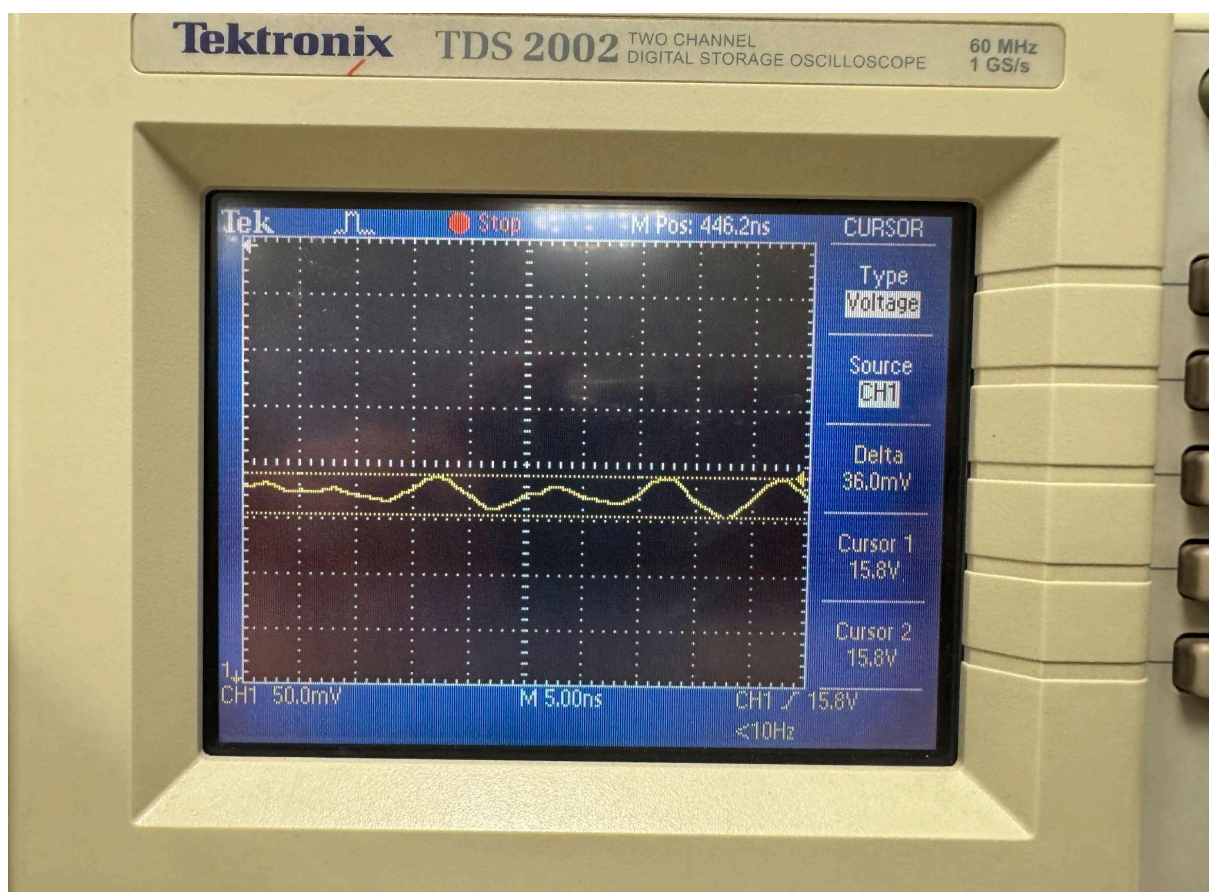


Figure 11. Ripple voltage

As we can see from the cursors, the ripple is very small **0.36V**, satisfying the criteria.

## Components Cost

| Component        | Type                   | Quantity | Unit Price (TL) | Total (TL)      |
|------------------|------------------------|----------|-----------------|-----------------|
| NE555            | Timer IC               | 1        | 6.49            | 6.49            |
| BC238            | NPN BJT                | 1        | 1               | 1               |
| BC308            | PNP BJT                | 1        | 1.07            | 1.07            |
| 1N5819           | Schottky Diode         | 4        | 1.06            | 4.24            |
| 100 $\mu$ F 25V  | Electrolytic Capacitor | 5        | 0.7             | 3.5             |
| 2.2nF            | Ceramic Capacitor      | 2        | 0.29            | 0.58            |
| 0.47 $\mu$ F 63V | Ceramic Capacitor      | 1        | 0.38            | 0.38            |
| 1/4W Resistors   | Standard Resistors     | 5        | 0.12            | 0.6             |
| Grand Total      |                        |          |                 | <b>17.86 TL</b> |

Table 1. Cost Table

| $V_{onl}$ (V)<br>no Load | $V_o$ (V) with $R_L$ | $V_r$ (mV) with<br>$R_L$ | $I_s$ with $R_L$ | Efficiency, $\eta$<br>(%) | Total<br>component<br>Cost (TL) |
|--------------------------|----------------------|--------------------------|------------------|---------------------------|---------------------------------|
| <b>17.9V</b>             | <b>16.2V</b>         | <b>36mV</b>              | <b>104mA</b>     | <b>75%</b>                | <b>17.86TL</b>                  |

Figure 12. Final result table